

Use Case 10: Infrastructure as Code

- **Scenario:** Automate infrastructure provisioning.
- **Tasks:**
 - Use CloudFormation to create a stack for a web application.
 - Include EC2, RDS, and an S3 bucket in the stack.
 - Update the stack to add a new feature.

Step 1: Define Requirements

1. The web application stack will include:
 - **EC2 instance:** For hosting the application.
 - **RDS instance:** For the database backend.
 - **S3 bucket:** For static file storage (e.g., images, documents).

aws

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Asia Pacific (Mumbai)

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IAM

EC2

VPC

S3

EC2 > Instances

Instances (2) Info

Last updated 2 minutes ago

Connect

Instance state

Actions

Launch instances

Find Instance by attribute or tag (case-sensitive)

All states

< 1 >

☐	Name	Instance ID	Instance state	Instanc...	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public
☐	WebAppInstance	i-001c8ce9d75e89...	Running	t2.micro	2/2 checks passed	View alarms +	ap-south-1b	ec2-65-1-92-212.ap-so...	65.1.9
☐	AppEC2Instance	i-0bf1f8c43c6fcfbef	Running	t2.micro	2/2 checks passed	View alarms +	ap-south-1b	ec2-52-66-246-97.ap-s...	52.66.

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IAM

EC2

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S3

Aurora and RDS > Databases > mywebappdb

mywebappdb

Modify

Actions

Summary

DB identifier
mywebappdb

CPU
2.89%

Status
Available

Class
db.t3.micro

Role
Instance

Current activity
0 Connections

Engine
MySQL Community

Region & AZ
ap-south-1a

Recommendations

Connectivity & security

Monitoring

Logs & events

Configuration

Zero-ETL integrations

Maintenance & backups

Data migrations - new

Tags

Recommendations

Connectivity & security

Endpoint & port

Endpoint
mywebappdb.c52u2g8y4aug.ap-south-1.rds.amazonaws.com

Port
3306

Networking

Availability Zone
ap-south-1a

VPC
vpc-021985fa9433044c5

Subnet group
default

Subnets
subnet-0520f599370b76482
subnet-0afca459511a96a5c

Security

VPC security groups
ApplInfraStack-AppSecurityGroup-PiLQnrc82PVW (sg-079f4490a16d3e8e3)

Active

Publicly accessible
Yes

Certificate authority
rds-ca-rsa2048-g1

Certificate authority date

aws

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IAM

EC2

VPC

S3

Amazon S3 > Buckets > infrastructure-bkt

infrastructure-bkt Info

Objects

Properties

Permissions

Metrics

Management

Access Points

Objects (0)

Copy S3 URI

Copy URL

Download

Open

Delete

Actions

Create folder

Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

< 1 >

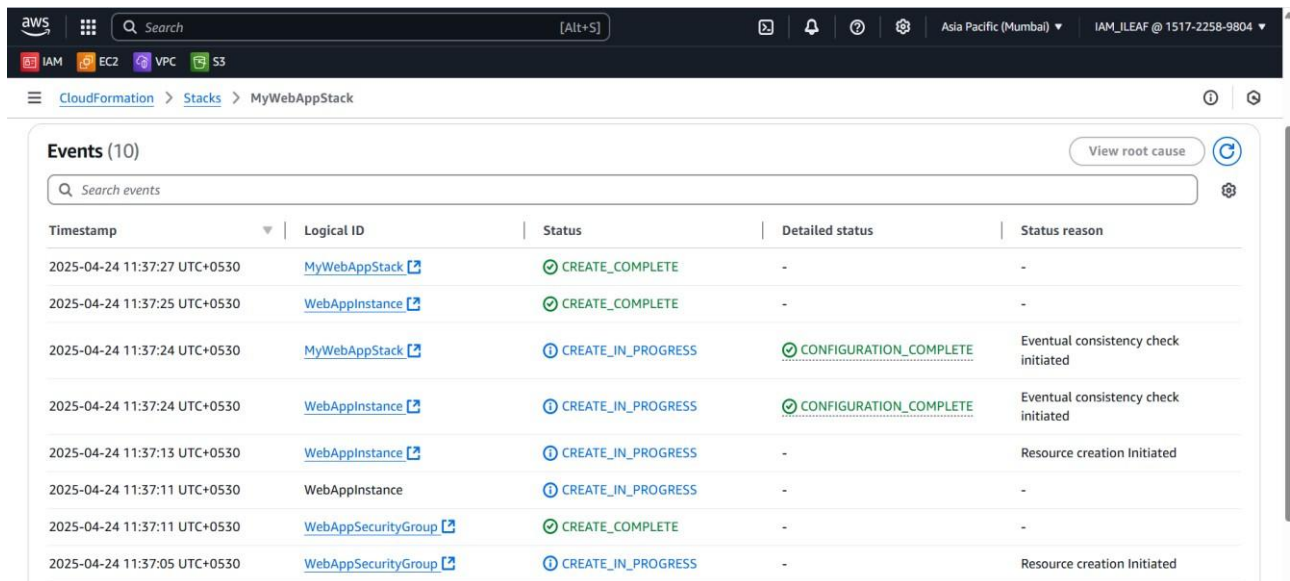
☐	Name	Type	Last modified	Size	Storage class
--------------------------	------	------	---------------	------	---------------

2. The stack should be secure, costeffective, and scalable.
3. Use **CloudFormation templates** to automate resource provisioning and ensure version control.

Step 2: Create a Basic CloudFormation Template

1. **Open CloudFormation Console:**
 - Navigate to the **AWS CloudFormation Console**.
2. **Write the Template:**
 - Use **YAML** or **JSON** to define your infrastructure.
3. **Save the Template:**
 - Save the file as `web-app-stack.yaml`.

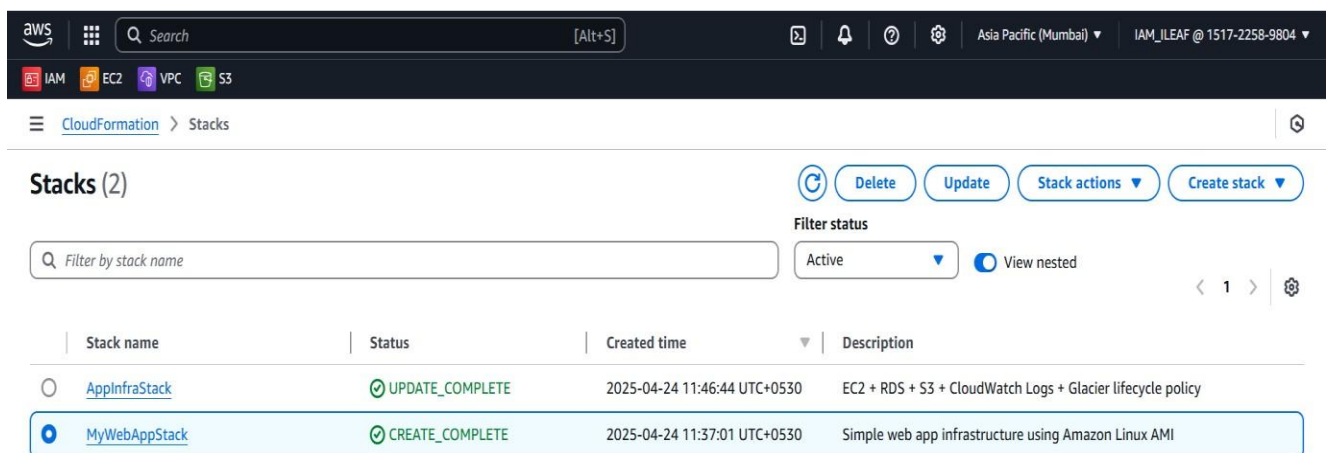
Step 3: Deploy the CloudFormation Stack



Events (10)

Search events

Timestamp	Logical ID	Status	Detailed status	Status reason
2025-04-24 11:37:27 UTC+0530	MyWebAppStack	CREATE_COMPLETE	-	-
2025-04-24 11:37:25 UTC+0530	WebAppInstance	CREATE_COMPLETE	-	-
2025-04-24 11:37:24 UTC+0530	MyWebAppStack	CREATE_IN_PROGRESS	CONFIGURATION_COMPLETE	Eventual consistency check initiated
2025-04-24 11:37:24 UTC+0530	WebAppInstance	CREATE_IN_PROGRESS	CONFIGURATION_COMPLETE	Eventual consistency check initiated
2025-04-24 11:37:13 UTC+0530	WebAppInstance	CREATE_IN_PROGRESS	-	Resource creation Initiated
2025-04-24 11:37:11 UTC+0530	WebAppInstance	CREATE_IN_PROGRESS	-	-
2025-04-24 11:37:11 UTC+0530	WebAppSecurityGroup	CREATE_COMPLETE	-	-
2025-04-24 11:37:05 UTC+0530	WebAppSecurityGroup	CREATE_IN_PROGRESS	-	Resource creation Initiated



Stacks (2)

Filter by stack name

Filter status: Active View nested

Stack name	Status	Created time	Description
AppInfraStack	UPDATE_COMPLETE	2025-04-24 11:46:44 UTC+0530	EC2 + RDS + S3 + CloudWatch Logs + Glacier lifecycle policy
MyWebAppStack	CREATE_COMPLETE	2025-04-24 11:37:01 UTC+0530	Simple web app infrastructure using Amazon Linux AMI

1. **Upload the Template:** In the CloudFormation Console, click **Create Stack** and choose **With New Resources**.

- Upload the web-app-stack.yaml file.

2. **Specify Parameters:** ◦ Enter required parameters like EC2 instance type and key pair.
3. **Review and Launch:** ◦ Verify the resources and permissions.
 - Click **Create Stack**.
4. **Monitor the Stack Creation:**
 - Wait for the stack status to reach `CREATE_COMPLETE`.

Step 4: Secure and Optimize the Stack

1. **Secure the EC2 Instance:**
 - Restrict **SSH access** to specific IPs in the WebAppSecurityGroup.
 - Enable **AWS Systems Manager Session Manager** for secure remote access without SSH:
 - Attach the

AmazonSSMManagedInstanceCore policy to the EC2 instance's IAM role.

2. **Secure the RDS Instance:**

- Use **KMS encryption** for the RDS storage.
- Ensure the RDS instance is in a **private subnet** for better security.

3. **Secure the S3 Bucket:**

- Block public access by enabling **Block Public Access Settings**.

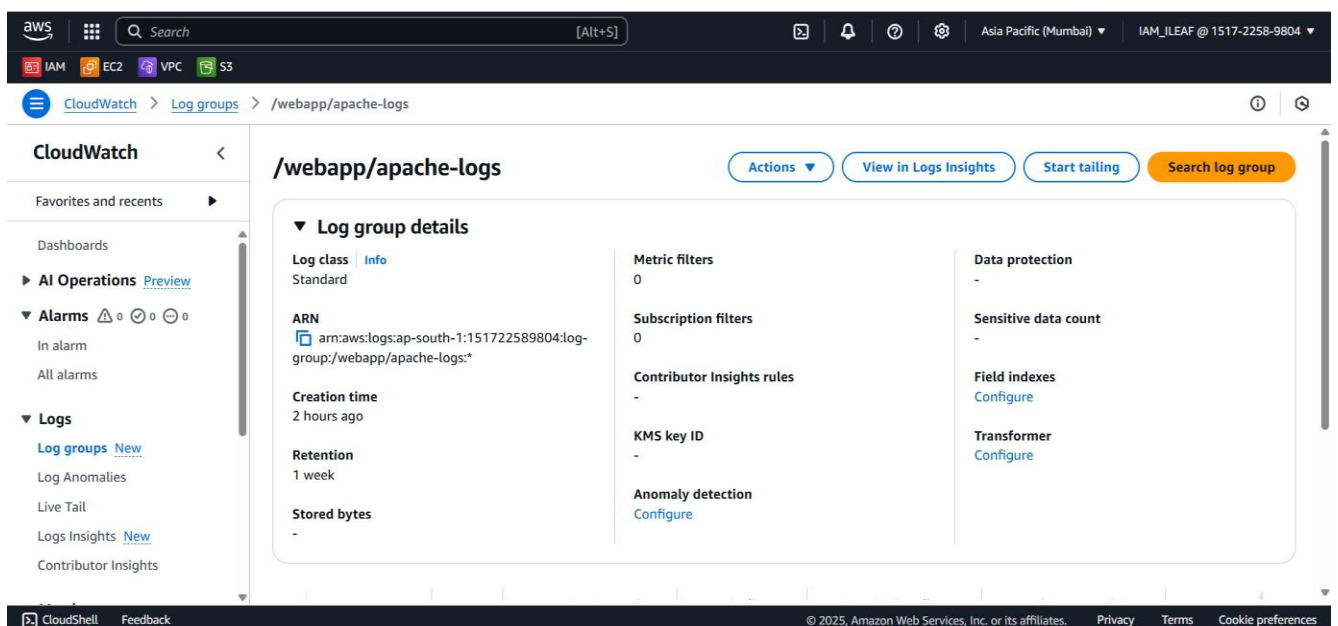
1. **Cost Optimization:**

- Use t3.micro instances for EC2 and RDS to stay within the **AWS Free Tier** (if applicable).
- Enable **Auto Scaling** for the EC2 instance to handle variable loads efficiently.
- Set up **lifecycle policies** to transition

old S3 objects to **Glacier**.

Step 5: Update the Stack to Add a New Feature

1. **New Feature:**
3. Add Cloudwatch
4. **Update the Stack:** ◦ In the CloudFormation Console, click on your stack and choose **Update**. ◦ Upload the modified template and deploy the update.
5. **Validate the Update:**



Lifecycle configuration

To manage your objects so that they are stored cost effectively throughout their lifecycle, configure their lifecycle. A lifecycle configuration is a set of rules that define actions that Amazon S3 applies to a group of objects. Lifecycle rules run once per day.

Default minimum object size for transitions
All storage classes 128K

Lifecycle rules (1)

Use lifecycle rules to define actions you want Amazon S3 to take during an object's lifetime such as transitioning objects to another storage class, archiving them, or deleting them after a specified period of time. [Learn more](#)

	Lifecycle rule name	Status	Scope	Current version acti...	Noncurrent versions...	Expired object delet...	Incomplete multip...
☐	TransitionToGlacier	Enabled	Entire bucket	Transition to Glacier Flexil	-	-	-

[View lifecycle configuration](#)

6.

- Test the ELB endpoint to confirm traffic is properly distributed to EC2 instances.

Step 6: Monitor and Maintain the Stack

1. Enable Logging:

- Use **CloudTrail** and **CloudWatch Logs** to track access and monitor resource usage.
- Set up alarms for unusual activity.

The screenshot shows the AWS CloudWatch console. The top navigation bar includes the AWS logo, a search bar, and various service icons. The left sidebar shows the 'CloudWatch' menu. The main content area is titled 'Log groups (1)' and includes a description: 'By default, we only load up to 10000 log groups.' Below this is a search bar and a table of log groups. The table has columns for 'Log group', 'Log class', 'Anomaly detection', 'Data', 'Sensitivity', 'Retention', and a checkbox. One log group is listed: '/webapp/apache-logs' with a retention of 1 week. The right sidebar contains buttons for 'Actions', 'View in Logs Insights', 'Start tailing', and 'Create log group'.

The screenshot shows the AWS CloudWatch console interface. At the top, the navigation bar includes the AWS logo, a search bar, and account information. The main header shows the path: CloudWatch > Log groups > /webapp/apache-logs. The left sidebar contains navigation links for CloudWatch, Log groups, and various log groups. The main content area displays the 'Log group details' for '/webapp/apache-logs'. This section includes a 'Log class' dropdown set to 'Standard', an 'ARN' field showing 'arn:aws:logs:ap-south-1:151722589804:log-group:/webapp/apache-logs*', and a 'Creation time' of '2 hours ago'. Other details include 'Retention' of '1 week', 'Stored bytes' of '-', 'Metric filters' of '0', 'Subscription filters' of '0', 'Contributor Insights rules' of '-', 'KMS key ID' of '-', and 'Anomaly detection' with a 'Configure' link. On the right, there are sections for 'Data protection', 'Sensitive data count', 'Field indexes' with a 'Configure' link, and 'Transformer' with a 'Configure' link. At the bottom, there are buttons for 'Actions', 'View in Logs Insights', 'Start tailing', and 'Search log group'.

CloudWatch

CloudWatch > Log groups > /webapp/apache-logs

/webapp/apache-logs

Log group details

Log class [Info](#)
Standard

ARN
[Info](#) arn:aws:logs:ap-south-1:151722589804:log-group:/webapp/apache-logs*

Creation time
2 hours ago

Retention
1 week

Stored bytes
-

Metric filters
0

Subscription filters
0

Contributor Insights rules
-

KMS key ID
-

Anomaly detection
[Configure](#)

Data protection
-

Sensitive data count
-

Field indexes
[Configure](#)

Transformer
[Configure](#)

[Actions](#) [View in Logs Insights](#) [Start tailing](#) [Search log group](#)

[Alt+J]

IAM
 EC2
 VPC
 S3

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Elastic IP addresses (1)

Find elastic IP addresses by attribute or tag

1

⚙️

☐	Name	Allocated IPv4 address	Type	Allocation ID	Reverse DNS record	Associated instance
☐	10-E-IP	3.6.226.233	Public IP	eipalloc-0194042373280df95	-	-

2. **Automate Backups:** ◦ Enable automated backups for RDS with a defined retention period.
 - Use **S3 Versioning** to preserve previous versions of objects.
3. **Optimize Costs:**
 - Use **Reserved Instances** or **Savings Plans** for predictable workloads.
 - Periodically review and delete unused resources.

```

1  {
2  "AWSTemplateFormatVersion": "2010-09-09",
3  "Description": "EC2 + RDS + S3 stack for web application",
4  "Resources": {
5  "AppSecurityGroup": {
6  "Type": "AWS::EC2::SecurityGroup",
7  "Properties": {
8  "GroupDescription": "Allow HTTP, SSH, and MySQL access",
9  "SecurityGroupIngress": [
10 { "IpProtocol": "tcp", "FromPort": 22, "ToPort": 22, "CidrIp": "0.0.0.0/0" },
11 { "IpProtocol": "tcp", "FromPort": 80, "ToPort": 80, "CidrIp": "0.0.0.0/0" },
12 { "IpProtocol": "tcp", "FromPort": 3306, "ToPort": 3306, "CidrIp": "0.0.0.0/0" }
13 ]
14 },
15 },
16 "AppEC2Instance": {
17 "Type": "AWS::EC2::Instance",
18 "Properties": {
19 "InstanceType": "t2.micro",
20 "ImageId": "ami-0f1dcc63b69a6438",
21 "SecurityGroups": [ { "Ref": "AppSecurityGroup" } ],
22 "UserData": {
23 "Fn::Base64": {
24 "Fn::Join": [
25 "",
26 [
27 "#!/bin/bash\n",
28 "yum update -y\n",
29 "yum install -y httpd mysql\n",
30 "systemctl start httpd\n",
31 "systemctl enable httpd\n",
32 "echo \"<h1>EC2 instance with RDS & S3</h1>\" > /var/www/html/index.html\n"
33 ]
34 }
35 }
36 }

```

```

4  "Resources": {
42 "AppRDSInstance": {
44 "Properties": {
45 "Engine": "mysql",
46 "EngineVersion": "8.0",
47 "MasterUsername": "adminuser",
48 "MasterUserPassword": "AdminPassword123!",
49 "DBInstanceIdentifier": "mywebappdb",
50 "VPCSecurityGroups": [ { "Fn::GetAtt": ["AppSecurityGroup", "GroupId"] } ],
51 "PubliclyAccessible": true,
52 "BackupRetentionPeriod": 1
53 },
54 },
55 },
56 },
57 "AppS3Bucket": {
58 "Type": "AWS::S3::Bucket",
59 "Properties": {
60 "BucketName": "infrastructure-bkt"
61 }
62 },
63 },
64 "Outputs": {
65 "EC2PublicIP": {
66 "Description": "Public IP of EC2 instance",
67 "Value": { "Fn::GetAtt": ["AppEC2Instance", "PublicIp"] }
68 },
69 "RDSHostname": {
70 "Description": "Endpoint of RDS instance",
71 "Value": { "Fn::GetAtt": ["AppRDSInstance", "Endpoint.Address"] }
72 },
73 "S3BucketName": {
74 "Description": "S3 bucket name",
75 "Value": { "Ref": "AppS3Bucket" }
76 }
77 }
78 }

```

```

task3-ec2-rds-s3-cloudwatchjson X
C:\Users\Arapal\Desktop> ILEAF > task3-ec2-rds-s3-cloudwatchjson > ...
1  {
2    "AWSTemplateFormatVersion": "2010-09-09",
3    "Description": "EC2 + RDS + S3 + CloudWatch Logs + Glacier lifecycle policy",
4    "Resources": {
5      "AppLogGroup": {
6        "Type": "AWS::Logs::LogGroup",
7        "Properties": {
8          "LogGroupName": "/webapp/apache-logs",
9          "RetentionInDays": 7
10       }
11     },
12     "AppInstanceRole": {
13       "Type": "AWS::IAM::Role",
14       "Properties": {
15         "AssumeRolePolicyDocument": {
16           "Version": "2012-10-17",
17           "Statement": [
18             {
19               "Effect": "Allow",
20               "Principal": {
21                 "Service": [ "ec2.amazonaws.com" ]
22               },
23               "Action": [ "sts:AssumeRole" ]
24             }
25           ]
26         },
27         "ManagedPolicyArns": [
28           "arn:aws:iam::aws:policy/CloudWatchAgentServerPolicy"
29         ]
30       }
31     },
32     "AppInstanceProfile": {
33       "Type": "AWS::IAM::InstanceProfile",
34       "Properties": {
35         "Roles": [ { "Ref": "AppInstanceRole" } ]
36       }

```

```

task3-ec2-rds-s3-cloudwatchjson X
C:\Users\Arapal\Desktop> ILEAF > task3-ec2-rds-s3-cloudwatchjson > ...
4    "Resources": {
32     "AppInstanceProfile": {
36       }
37     },
38     "AppSecurityGroup": {
39       "Type": "AWS::EC2::SecurityGroup",
40       "Properties": {
41         "GroupDescription": "Allow HTTP, SSH, and MySQL access",
42         "SecurityGroupIngress": [
43           { "IpProtocol": "tcp", "FromPort": 22, "ToPort": 22, "CidrIp": "0.0.0.0/0" },
44           { "IpProtocol": "tcp", "FromPort": 80, "ToPort": 80, "CidrIp": "0.0.0.0/0" },
45           { "IpProtocol": "tcp", "FromPort": 3306, "ToPort": 3306, "CidrIp": "0.0.0.0/0" }
46         ]
47       }
48     },
49     "AppEC2Instance": {
50       "Type": "AWS::EC2::Instance",
51       "Properties": {
52         "InstanceType": "t2.micro",
53         "ImageId": "ami-0f1dcc636b69a6438",
54         "IamInstanceProfile": { "Ref": "AppInstanceProfile" },
55         "SecurityGroups": [ { "Ref": "AppSecurityGroup" } ],
56         "UserData": {
57           "Fn::Base64": {
58             "Fn::Join": [
59               "",
60               [
61                 "#!/bin/bash\n",
62                 "yum update -y\n",
63                 "yum install -y httpd amazon-cloudwatch-agent\n",
64                 "systemctl start httpd\n",
65                 "systemctl enable httpd\n",
66                 "echo \"<h1>EC2 with RDS, S3 and CloudWatch Logs</h1>\" > /var/www/html/index.html\n",
67                 "cat <<EOF > /opt/aws/amazon-cloudwatch-agent/etc/amazon-cloudwatch-agent.json\n",
68                 "{\n",

```

```

67         "cat <<EOF > /opt/aws/amazon-cloudwatch-agent/etc/amazon-cloudwatch-agent.json\n",
68         "{\n",
69         "  \"logs\": {\n",
70         "    \"logs_collected\": {\n",
71         "      \"files\": {\n",
72         "        \"collect_list\": [\n",
73         "          {\n",
74         "            \"file_path\": \"\n/var/log/httpd/access_log\n\",\n",
75         "            \"log_group_name\": \"\n/webapp/apache-logs\n\",\n",
76         "            \"log_stream_name\": \"\naccess_log\n\",\n",
77         "          },\n",
78         "          {\n",
79         "            \"file_path\": \"\n/var/log/httpd/error_log\n\",\n",
80         "            \"log_group_name\": \"\n/webapp/apache-logs\n\",\n",
81         "            \"log_stream_name\": \"\nerror_log\n\",\n",
82         "          }\n",
83         "        ]\n",
84         "      }\n",
85         "    }\n",
86         "  }\n",
87         "EOF\n",
88         "/opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-ctl -a fetch-config -m ec2 -c file:/opt/aws/amazon-cloudwatch-agent/etc/amazon-clo
89     ]\n",
90     }\n",
91   },\n",
92   \"Tags\": [\n",
93     {\n",
94       \"Key\": \"Name\", \"Value\": \"AppEC2Instance\" }\n",
95     ]\n",
96   ]\n",

```

```

3     },\n",
4     \"Tags\": [\n",
5       {\n",
6         \"Key\": \"Name\", \"Value\": \"AppEC2Instance\" }\n",
7       ]\n",
8     ],\n",
9     \"AppRDSInstance\": {\n",
10       \"Type\": \"AWS::RDS::DBInstance\",\n",
11       \"Properties\": {\n",
12         \"AllocatedStorage\": \"20\",\n",
13         \"DBInstanceClass\": \"db.t3.micro\",\n",
14         \"Engine\": \"MySQL\",\n",
15         \"EngineVersion\": \"8.0\",\n",
16         \"MasterUsername\": \"adminuser\",\n",
17         \"MasterUserPassword\": \"AdminPassword123!\",\n",
18         \"DBInstanceIdentifier\": \"mywebappdb\",\n",
19         \"VPCSecurityGroups\": [ {\n",
20           \"Fn::GetAtt\": [ \"AppSecurityGroup\", \"GroupId\" ] }\n",
21         ],\n",
22         \"PubliclyAccessible\": true,\n",
23         \"BackupRetentionPeriod\": 1\n",
24       }\n",
25     },\n",
26     \"AppS3Bucket\": {\n",
27       \"Type\": \"AWS::S3::Bucket\",\n",
28       \"Properties\": {\n",
29         \"BucketName\": \"infrastructure-bkt\",\n",
30         \"LifecycleConfiguration\": {\n",
31           \"Rules\": [\n",
32             {\n",
33               \"Id\": \"TransitionToGlacier\",\n",
34               \"Status\": \"Enabled\",\n",
35               \"Prefix\": \"\",\n",
36               \"Transitions\": [\n",
37                 {\n",
38                   \"Days\": 30,\n",
39                   \"StorageClass\": \"GLACIER\"\n",
40                 }\n",
41               ]\n",
42             }\n",
43           ]\n",
44         }\n",
45       }\n",
46     }\n",
47   ]\n",

```

```
129     ],
130     }
131   ]
132   }
133   }
134   }
135 },
136 "Outputs": {
137   "EC2PublicIP": {
138     "Description": "Public IP of EC2 instance",
139     "Value": { "Fn::GetAtt": ["AppEC2Instance", "PublicIp"] }
140   },
141   "RDSHostname": {
142     "Description": "Endpoint of RDS instance",
143     "Value": { "Fn::GetAtt": ["AppRDSInstance", "Endpoint.Address"] }
144   },
145   "S3BucketName": {
146     "Description": "S3 bucket name",
147     "Value": { "Ref": "AppS3Bucket" }
148   },
149   "CloudWatchLogGroup": {
150     "Description": "Log group for Apache logs",
151     "Value": "/webapp/apache-logs"
152   }
153 }
154 }
155
```